

高等微积分 (2)

第十四章: 多变量函数的微分学

习题课

提要: 典型例题等

例 1. 设 $f: (0, +\infty) \rightarrow \mathbb{R}$ 为二阶连续可导函数
且 $u(x, y, z) = f(\sqrt{x^2 + y^2 + z^2})$ 满足

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0,$$

求函数 f 的表达式.

解: 由题设可知 $\frac{\partial u}{\partial x} = \frac{xf'(\sqrt{x^2+y^2+z^2})}{\sqrt{x^2+y^2+z^2}}$, 则

$$\frac{\partial^2 u}{\partial x^2} = \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{x^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} - \frac{x^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}.$$

由对称性可得

$$\frac{\partial^2 u}{\partial y^2} = \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{y^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} - \frac{y^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}},$$

$$\frac{\partial^2 u}{\partial z^2} = \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{z^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} - \frac{z^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}},$$

于是我们有

$$\begin{aligned} 0 &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{x^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} \\ &\quad - \frac{x^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{y^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} \\ &\quad - \frac{y^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{z^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} \\ &\quad - \frac{z^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} = \frac{2f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + f''(\sqrt{x^2 + y^2 + z^2}). \end{aligned}$$

从而 $\forall r > 0, f''(r) + \frac{2f'(r)}{r} = 0$, 故 $f'(r) = -\frac{C_1}{r^2}$,
进而 $f(r) = \frac{C_1}{r} + C_2$, 其中 C_1, C_2 为常数.

例 2. 设 $u = x^2 + y^2 + z^2$, 其中 $z = z(x, y)$ 是由方程 $z = x - ye^z$ 确定的隐函数, 求 du 和 $\frac{\partial^2 u}{\partial x \partial y}$.

解: 由题设知 $\frac{\partial u}{\partial x} = 2x + 2z \frac{\partial z}{\partial x}$, $\frac{\partial u}{\partial y} = 2y + 2z \frac{\partial z}{\partial y}$.

注意到 $z = x - ye^z$, 两边分别对 x, y 求偏导数可得 $\frac{\partial z}{\partial x} = \frac{1}{1+ye^z}$, $\frac{\partial z}{\partial y} = -\frac{e^z}{1+ye^z}$. 于是

$$du = \left(2x + \frac{2z}{1+ye^z}\right)dx + \left(2y - \frac{2ze^z}{1+ye^z}\right)dy.$$

与此同时, 我们也有

$$\begin{aligned}\frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(2y + 2z \frac{\partial z}{\partial y} \right) = 2 \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} + 2z \frac{\partial^2 z}{\partial x \partial y} \\ &= -\frac{2e^z}{(1 + ye^z)^2} + 2z \frac{\partial}{\partial x} \left(-\frac{e^z}{1 + ye^z} \right) \\ &= -\frac{2e^z}{(1 + ye^z)^2} - 2z \cdot \frac{e^z(1 + ye^z) \frac{\partial z}{\partial x} - e^z \cdot ye^z \frac{\partial z}{\partial x}}{(1 + ye^z)^2}\end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(2y + 2z \frac{\partial z}{\partial y} \right) \\
&= -\frac{2e^z}{(1 + ye^z)^2} - \frac{2ze^z}{(1 + ye^z)^3} \\
&= -\frac{2e^z(1 + ye^z + z)}{(1 + ye^z)^3} = -\frac{2(1 + x)e^z}{(1 + ye^z)^3}.
\end{aligned}$$

例 3. 假设 $z = z(x, y)$ 为二阶连续可导且满足

$$A \frac{\partial^2 z}{\partial x^2} + 2B \frac{\partial^2 z}{\partial x \partial y} + C \frac{\partial^2 z}{\partial y^2} = 0,$$

其中 $B^2 - AC > 0$ 且 $C \neq 0$. 若令

$$\begin{cases} u = x + \alpha y, \\ v = x + \beta y, \end{cases}$$

试确定 α, β 为何值时能使原方程变为

$$\frac{\partial^2 z}{\partial u \partial v} = 0.$$

解: 由题设可知

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) z,$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right)^2 z,$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \alpha + \frac{\partial z}{\partial v} \cdot \beta = \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right) z,$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right)^2 z,$$

$$\begin{aligned}\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right) z \\ &= \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right) z.\end{aligned}$$

带入题设方程可得

$$\begin{aligned} 0 &= A \frac{\partial^2 z}{\partial x^2} + 2B \frac{\partial^2 z}{\partial x \partial y} + C \frac{\partial^2 z}{\partial y^2} \\ &= A \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right)^2 z + 2B \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right) z \\ &\quad + C \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right)^2 z \\ &= (A + 2B\alpha + C\alpha^2) \frac{\partial^2 z}{\partial u^2} \\ &\quad + 2(A + B(\alpha + \beta) + C\alpha\beta) \frac{\partial^2 z}{\partial u \partial v} \end{aligned}$$

于是题设方程能变成 $\frac{\partial^2 z}{\partial u \partial v} = 0$ 当且仅当

$$A + 2B\alpha + C\alpha^2 = 0, \quad A + 2B\beta + C\beta^2 = 0,$$

$$A + B(\alpha + \beta) + C\alpha\beta \neq 0.$$

由于 $B^2 - AC > 0$, 因此我们有

$$\alpha = \frac{-B \mp \sqrt{B^2 - AC}}{C},$$

$$\beta = \frac{-B \pm \sqrt{B^2 - AC}}{C}.$$

此时我们还有

$$\begin{aligned} & A + B(\alpha + \beta) + C\alpha\beta \\ &= A - \frac{2B^2}{C} + A = \frac{2}{C}(AC - B^2) \neq 0. \end{aligned}$$

于是题设方程能变成 $\frac{\partial^2 z}{\partial u \partial v} = 0$ 当且仅当

$$\alpha = \frac{-B \mp \sqrt{B^2 - AC}}{C}, \quad \beta = \frac{-B \pm \sqrt{B^2 - AC}}{C}.$$

此时存在两个连续可导函数 f, g 使得

$$z(x, y) = f(u) + g(v) = f(x + \alpha y) + g(x + \beta y).$$

例 4. 设 $x = f(y, z)$, $y = g(x, z)$, $z = h(x, y)$,
其中 f, g, h 为可微函数. 求证:

$$\frac{\partial f}{\partial y} \cdot \frac{\partial g}{\partial z} \cdot \frac{\partial h}{\partial x} = -1.$$

证明: 由题设可知

$$x = f(g(x, z), z), \quad z = h(x, g(x, z)).$$

由此立刻可得

$$1 = \frac{\partial f}{\partial y} \cdot \frac{\partial g}{\partial x}, \quad 0 = \frac{\partial h}{\partial x} + \frac{\partial h}{\partial y} \cdot \frac{\partial g}{\partial x}, \quad 1 = \frac{\partial h}{\partial y} \cdot \frac{\partial g}{\partial z}.$$

于是我们就有

$$\frac{\partial f}{\partial y} = \frac{1}{\frac{\partial g}{\partial x}}, \quad \frac{\partial h}{\partial x} = -\frac{\partial h}{\partial y} \cdot \frac{\partial g}{\partial x}, \quad \frac{\partial g}{\partial z} \cdot \frac{\partial h}{\partial y} = 1.$$

进而立刻可得知

$$\begin{aligned} \frac{\partial f}{\partial y} \cdot \frac{\partial g}{\partial z} \cdot \frac{\partial h}{\partial x} &= \frac{1}{\frac{\partial g}{\partial x}} \cdot \frac{\partial g}{\partial z} \cdot \left(-\frac{\partial h}{\partial y} \frac{\partial g}{\partial x} \right) \\ &= -\frac{\partial g}{\partial z} \cdot \frac{\partial h}{\partial y} = -1. \end{aligned}$$

例 5. $\forall (x, y, z) \in \mathbb{R}^3$, 定义

$$f(x, y, z) = x^2 + 2y - xyz.$$

设 f 在点 $(1, 1, 0)$ 处的梯度方向为 $\vec{\ell}$, 求 $\frac{\partial f}{\partial \ell}(1, 1, 0)$.

解: 由题设可知

$$\text{grad} f(1, 1, 0) = \begin{pmatrix} 2x - yz \\ 2 - xz \\ -xy \end{pmatrix} \Big|_{(1,1,0)} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix},$$

于是我们有 $\frac{\partial f}{\partial \ell}(1, 1, 0) = \|\text{grad} f(1, 1, 0)\| = 3$.

例 6. 设隐函数 $z = z(x, y)$ 由方程

$$z = f(x + y + z)$$

确定, 其中 f 为 $\mathcal{C}^{(2)}$ 类函数且 $f' \neq 1$, 求 $\frac{\partial^2 z}{\partial x^2}$.

解: 将方程 $z = f(x + y + z)$ 对 x 求偏导数, 则

$$\begin{aligned}\frac{\partial z}{\partial x} &= f'(x + y + z) \frac{\partial(x + y + z)}{\partial x} \\ &= f'(x + y + z) \left(1 + \frac{\partial z}{\partial x}\right),\end{aligned}$$

由此立刻可得 $\frac{\partial z}{\partial x} = \frac{f'(x+y+z)}{1-f'(x+y+z)}$.

进而我们就有

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{f'(x+y+z)}{1-f'(x+y+z)} \right) \\ &= \frac{\partial}{\partial x} \left(-1 + \frac{1}{1-f'(x+y+z)} \right) \\ &= -\frac{1}{(1-f'(x+y+z))^2} \cdot \frac{\partial(1-f'(x+y+z))}{\partial x}\end{aligned}$$

$$\begin{aligned}
&= \frac{f''(x+y+z)}{(1-f'(x+y+z))^2} \cdot \frac{\partial(x+y+z)}{\partial x} \\
&= \frac{f''(x+y+z)}{(1-f'(x+y+z))^2} \left(1 + \frac{\partial z}{\partial x}\right) \\
&= \frac{f''(x+y+z)}{(1-f'(x+y+z))^3}.
\end{aligned}$$

例 7. 设 p 为实数. $\forall (x, y) \in \mathbb{R}^2$, 令

$$f(x, y) = \begin{cases} (x^2 + y^2)^p \sin \frac{1}{\sqrt{x^2 + y^2}}, & \text{若 } (x, y) \neq (0, 0), \\ 0, & \text{若 } (x, y) = (0, 0). \end{cases}$$

请分析 p 取何值时, 函数 f 在原点处:

(1) 连续; (2) 可导; (3) 可微.

解: (1) 若 $p > 0$, 则 $\forall (x, y) \in \mathbb{R}^2$, 我们有

$$|f(x, y)| \leq (x^2 + y^2)^p.$$

由夹逼原理可得 $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0 = f(0, 0)$. 连续.

现假设 $p \leq 0$. $\forall n \geq 1$, 令 $x_n = \frac{1}{2n\pi}$, $y_n = \frac{1}{2n\pi + \frac{\pi}{2}}$,

则 $f(x_n, 0) = 0$, $f(y_n, 0) = y_n^{2p} \geq 1$. 故 $\{x_n\}$, $\{y_n\}$

收敛到 0, 但 $\{f(x_n, 0)\}$, $\{f(y_n, 0)\}$ 却不收敛到同一个极限. 这表明 f 在原点间断.

综上所述可知 f 在原点连续当且仅当 $p > 0$.

(2) 若 $p > \frac{1}{2}$, 由夹逼原理可知

$$\lim_{x \rightarrow 0} \frac{f(x,0)-f(0,0)}{x} = \lim_{x \rightarrow 0} |x|^{2p-1}(\operatorname{sgn} x) \sin \frac{1}{|x|} = 0,$$

$$\lim_{y \rightarrow 0} \frac{f(0,y)-f(0,0)}{y} = \lim_{y \rightarrow 0} |y|^{2p-1}(\operatorname{sgn} y) \sin \frac{1}{|y|} = 0,$$

故 f 在原点可导且 $f'_x(0,0) = f'_y(0,0) = 0$.

若 $p \leq \frac{1}{2}$, 由 $\lim_{x \rightarrow 0} (\operatorname{sgn} x) \sin \frac{1}{|x|}$ 不存在可知 f 在原点不可导. 故 f 在原点可导当且仅当 $p > \frac{1}{2}$.

(3) 若 f 在原点可微, 则它在该点可导, 故 $p > \frac{1}{2}$.

现假设 $p > \frac{1}{2}$, 此时我们有

$$\begin{aligned} & \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y}{\sqrt{x^2 + y^2}} \\ &= \lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2)^{p-\frac{1}{2}} \sin \frac{1}{\sqrt{x^2 + y^2}} = 0, \end{aligned}$$

于是由可微的定义可知 f 在原点可微.

综上所述可知 f 在原点可微当且仅当 $p > \frac{1}{2}$.

例 8. 固定 $k > 0$. $\forall (x, y) \in \mathbb{R}^2$, 定义

$$f(x, y) = \begin{cases} \frac{|xy|^k}{x^2+y^2}, & \text{若 } (x, y) \neq (0, 0), \\ 0, & \text{若 } (x, y) = (0, 0). \end{cases}$$

问 k 为何值时函数 f 在原点连续, 可导, 可微或连续可导?

解: 连续性. 当 $k > 1$ 时, $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$,

$$\frac{|xy|^k}{x^2+y^2} \leq \frac{(\frac{1}{2}(x^2+y^2))^k}{x^2+y^2} = \frac{1}{2^k}(x^2+y^2)^{k-1}.$$

于是由夹逼原理可知此时函数 f 在原点连续.

当 $k = 1$ 时, 由于 $\lim_{x \rightarrow 0} f(x, x) = \frac{1}{2}$, 由复合极限法则可知此时函数 f 在原点不连续.

当 $k < 1$ 时, 因 $\lim_{x \rightarrow 0} f(x, x) = \lim_{x \rightarrow 0} \frac{1}{2}|x|^{2k-2} = +\infty$, 则由复合极限法则可知 f 在原点不连续.

综上所述可知 f 在原点连续当且仅当 $k > 1$.

可导性. $\forall x, y \in \mathbb{R} \setminus \{0\}$, 因 $f(x, 0) = f(0, y) = 0$, 于是由偏导数的定义可知 $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) = 0$, 故函数 f 在原点处可导.

可微性. 因 $\frac{\partial f}{\partial x}(0,0) = \frac{\partial f}{\partial y}(0,0) = 0$, 则由微分的定义立刻可知 f 在原点可微当且仅当

$$\lim_{(x,y) \rightarrow (0,0)} \frac{|xy|^k}{(x^2 + y^2)^{\frac{3}{2}}} = 0,$$

而这又等价于说

$$\lim_{(x,y) \rightarrow (0,0)} \frac{|xy|^{\frac{2k}{3}}}{x^2 + y^2} = 0,$$

由此可知 f 在原点处可微当且仅当 $k > \frac{3}{2}$.

连续可导性. 若 f 在原点连续可导, 则由前面讨论可知 $k > \frac{3}{2}$. 此时 $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$,

$$\frac{\partial f}{\partial x}(x, y) = \frac{k|x|^{k-1}|y|^k \operatorname{sgn} x}{x^2 + y^2} - \frac{2x|xy|^k}{(x^2 + y^2)^2},$$

$$\frac{\partial f}{\partial y}(x, y) = \frac{k|x|^k|y|^{k-1} \operatorname{sgn} y}{x^2 + y^2} - \frac{2y|xy|^k}{(x^2 + y^2)^2}.$$

由此立刻可得

$$\begin{aligned} \left| \frac{\partial f}{\partial x}(x, y) \right| &\leq \frac{k|x|^{k-1}|y|^k}{x^2 + y^2} + \frac{2|x|^{k+1}|y|^k}{(x^2 + y^2)^2} \\ &\leq \frac{k}{2^{k-1}}(x^2 + y^2)^{k-2}|y| + \frac{1}{2^{k-1}}(x^2 + y^2)^{k-2}|x| \\ &\leq \frac{k}{2^{k-1}}(x^2 + y^2)^{k-\frac{3}{2}} + \frac{1}{2^{k-1}}(x^2 + y^2)^{k-\frac{3}{2}} \\ &= \frac{k+1}{2^{k-1}}(x^2 + y^2)^{k-\frac{3}{2}}, \\ \left| \frac{\partial f}{\partial y}(x, y) \right| &\leq \frac{k|x|^k|y|^{k-1}}{x^2 + y^2} + \frac{2|x|^k|y|^{k+1}}{(x^2 + y^2)^2} \leq \frac{k+1}{2^{k-1}}(x^2 + y^2)^{k-\frac{3}{2}}, \end{aligned}$$

例 9. 假设由方程

$$f(u^2 - x^2, u^2 - y^2, u^2 - z^2) = 0$$

可确定 $u = \varphi(x, y, z)$, 其中 f, φ 可微. 记

$$X = (u^2 - x^2, u^2 - y^2, u^2 - z^2).$$

若 $xyz u \neq 0$ 且 $\partial_1 f(X) + \partial_2 f(X) + \partial_3 f(X) \neq 0$,

求证: $\frac{u'_x}{x} + \frac{u'_y}{y} + \frac{u'_z}{z} = \frac{1}{u}$.

证明: 由于 $f(u^2 - x^2, u^2 - y^2, u^2 - z^2) = 0$, 故

$$\begin{aligned} \partial_1 f(X) \frac{\partial}{\partial x} (u^2 - x^2) + \partial_2 f(X) \frac{\partial}{\partial x} (u^2 - y^2) \\ + \partial_3 f(X) \frac{\partial}{\partial x} (u^2 - z^2) = 0, \end{aligned}$$

也即我们有

$$\partial_1 f(X)(2uu'_x - 2x) + \partial_2 f(X)(2uu'_x) + \partial_3 f(X)(2uu'_x) = 0,$$

$$\text{故 } \frac{u'_x}{x} (\partial_1 f(X) + \partial_2 f(X) + \partial_3 f(X)) = \frac{1}{u} \partial_1 f(X).$$

由对称性立刻可得

$$\frac{u'_y}{y}(\partial_1 f(X) + \partial_2 f(X) + \partial_3 f(X)) = \frac{1}{u} \partial_2 f(X),$$
$$\frac{u'_z}{z}(\partial_1 f(X) + \partial_2 f(X) + \partial_3 f(X)) = \frac{1}{u} \partial_3 f(X).$$

于是我们就有

$$\frac{u'_x}{x} + \frac{u'_y}{y} + \frac{u'_z}{z} = \frac{1}{u}.$$

例 10. $\forall (x, y) \in \mathbb{R}^2$, 定义 $f(x, y) = \sqrt{x^2 + y^4}$.

研究 f 在原点的连续性, 可导性以及可微性.

解: 由于 f 为初等函数, 因此在原点连续.

与此同时, 由于极限

$$\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{|x|}{x}$$

不存在, 故 f 在原点处没有关于第一个变量的偏导数, 进而可知 f 在原点处不可微.

例 11. 假设 $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ 为二阶连续可导函数, 而隐函数 $z = z(x, y)$ 可由方程 $x + y = f(x, z)$ 确定, 其中 $\partial_2 f(x, z) \neq 0$. 计算 $\frac{\partial^2 z}{\partial x \partial y}$.

解: 将方程两边分别对 x, y 求偏导可得

$$1 = \partial_1 f(x, z) + \partial_2 f(x, z) \frac{\partial z}{\partial x}, \quad 1 = \partial_2 f(x, z) \frac{\partial z}{\partial y},$$

由此我们立刻可知

$$\frac{\partial z}{\partial x} = \frac{1 - \partial_1 f(x, z)}{\partial_2 f(x, z)}, \quad \frac{\partial z}{\partial y} = \frac{1}{\partial_2 f(x, z)}.$$

于是我们有

$$\begin{aligned}\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{1}{\partial_2 f(x, z)} \right) = - \frac{1}{(\partial_2 f(x, z))^2} \frac{\partial}{\partial x} (\partial_2 f(x, z)) \\&= - \frac{1}{(\partial_2 f(x, z))^2} \left(\partial_{12} f(x, z) + \partial_{22} f(x, z) \frac{\partial z}{\partial x} \right) \\&= - \frac{1}{(\partial_2 f(x, z))^2} \left(\partial_{12} f(x, z) + \partial_{22} f(x, z) \cdot \frac{1 - \partial_1 f(x, z)}{\partial_2 f(x, z)} \right) \\&= \frac{\partial_{22} f(x, z) \cdot \partial_1 f(x, z) - \partial_{12} f(x, z) \cdot \partial_2 f(x, z) - \partial_{22} f(x, z)}{(\partial_2 f(x, z))^3}\end{aligned}$$

同学们辛苦了!